

DATA MINING PROJECT

Master in Data Science and Advanced Analytics

NOVA Information Management School

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Amazing International Airlines Inc. (AIAI)

Group 51

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1. ABSTRACT

This exploratory data analysis (EDA) examines three years of loyalty and flight data for Amazing International Airlines Inc. (AIAI) to support a new customer segmentation strategy. The analysis revealed seasonal travel peaks, year-over-year growth, and highly skewed customer data. However, we also uncovered significant data integrity issues, including duplicate records, inconsistent point calculations, and negative balances, all of which require corrective preprocessing. These findings provide a necessary foundation for future clustering to help AIAI tailor marketing strategies to distinct traveler profiles.

2. INTRODUCTION

Amazing International Airlines Inc. (AIAI) operates in one of the most competitive service industries, where customer loyalty and personalized experiences have become decisive factors for business sustainability. The airline industry faces increasing price competition, fluctuating demand, and growing expectations for easy, personalized travel experiences. Within this context, AIAI seeks to strengthen its customer retention strategy through a data-driven understanding of its loyalty program members.

AIAI aims to strengthen customer loyalty and profitability by developing a data-driven customer segmentation strategy. This segmentation initiative originates from a pedagogical impetus embedded in data-centric inquiry. Through the analysis of three years of loyalty and flight activity data, the company seeks to identify distinct customer groups with different behavioral, value-based, and demographic characteristics. These insights will inform personalized marketing, loyalty rewards optimization, and customer retention initiatives.

2.1. Business Understanding

AIAI runs a multi-tier loyalty program designed to reward its most frequent travelers. Although the company collects a large amount of data on flights and customer interactions, it still lacks a clear, structured understanding of the different types of customers it serves. This segmentation project is focused on using data more effectively.

By applying data mining techniques, AIAI hopes to identify hidden patterns in its customer database. Distinguishing, for example, loyal, high-value travelers from occasional flyers or inactive members. The main challenge is turning raw data into meaningful insights that can drive personalized marketing, smarter loyalty rewards, and more effective retention strategies. Gaining a better grasp of how customers differ in flight habits, travel distance, and loyalty behavior will allow AIAI to offer more tailored experiences and strengthen its competitive edge.

This stage aligns with the *Business Understanding* phase of the CRISP-DM methodology, which defines the strategic goals and analytical direction of the project.

Table 1: CRISP-DM Business Understanding

Phase	Activity	Work Product
Business Understanding	Determine business goals	- Segment customers by economic value. - Analyze purchasing and travel behaviors. - Identify demographic differences for personalization.
	Assess situation	Highly competitive airline industry
	Determine data mining goals	Customer segmentation for personalized marketing
	Produce Project Plan	Initial Evaluation: Possibility of further applying techniques for segmentation such as K-means, Hierarchical Clustering, DBSCAN, PCA. Project Plan: Management in <i>ClickUp</i>

3. DATA ANALYSIS

Table 2: CRISP-DM Data Analysis

Data Analysis Phase	Activity
Data understanding	Collect initial data
	Describe data
	Explore data
	Verify data quality

In alignment with the CRISP-DM Data Understanding phase, we began by exploring the **flights dataset** to understand feature behaviors. The data spans 36 months (2019-2021) with no initial missing values. A statistical review highlighted highly skewed distributions, which are key for clustering. For *NumFlights*, a median of 0 indicates that most customers have months of inactivity, while a small group of frequent flyers skews the average. Similarly, “*DistanceKM*” shows high variability, reflecting a diverse customer base of both short and long-haul travelers. This pattern continues with loyalty points: most records show zero redemption, suggesting that customers are primarily accumulators rather than redeemers.

A peculiar anomaly was found exclusively in the 2019 data, where flight numbers often appeared as fractions, unlike the whole numbers seen in 2020 and 2021. Our analysis also revealed clear trends: travel activity showed strong seasonality, peaking in the summer months and flight activity grew steadily year-over-year.

We confirmed that two pairs of variables were perfectly redundant. *DistanceKM* measured the same thing as *PointsAccumulated*, and *PointsRedeemed* was identical to *DollarCostPointsRedeemed*. This means one variable from each pair must be dropped to avoid over-weighting that information in our analysis.

Finally, we identified several critical data errors that must be fixed. We found over 11,700 duplicate rows, likely from a system aggregation error. We also found 5,858 records that showed distance traveled and point activity but incorrectly listed zero flights. Furthermore, the points system data was inconsistent, as point calculations didn't always follow the business rules, and 1,900 customers had unexplained negative point balances.

The **customer dataset** consists of information from 16,757 customers and 20 variables, encompassing demographic, geographic, socioeconomic and loyalty-related information. The dataset exclusively includes customers from Canada, making the country variable a categorical constant, however other geographical features retain analytical relevance that may enable future spatial regional assessments. While this limits cross-country comparisons, other geographical attributes such as province, city, latitude, and longitude remain analytically relevant, allowing for regional and spatial assessments within the Canadian market. In exploratory analysis this is reinforced by *Latitude* and *Longitude* multimodal patterns observed suggest the existence of distinct geographic market segments.

An assessment of missing values reveals a robust dataset with only three features with missing values. Two variables with partial data absence, being *Income* and *Customer Lifetime Value*, having 20 missing entries (1,2% of records) and *CancellationDate* displaying a substantial proportion of missing values (86% of records), which can mainly correspond to customers with ongoing memberships.

The descriptive analysis of key numerical variables indicates substantial variability in both *Income* and *customer lifetime value*, each having a right-skewed nature, which indicates the presence of a small number of high-income and high-value customers who significantly influence the averages, which can be relevant for further results specifically in targeting marketing campaigns.

4. RESULTS

Data Analysis Phase	Activity
Data preparation	Select and clear data
	Construct data
	Integrate data
	Format data

4.1. Changing Values

When we were conducting the initial analysis of our customer data, while checking for the number of entries of each column (checking if there were any missing values) and also checking for the data types, we realized that the columns *Enrollment Date Opening* and *CancellationDate* were defined with the

data type “object”. In order to be able to do further analysis of these columns, we changed the data type of both columns to “datetime64[ns]”.

In the flights dataset, we first resolved 11,772 duplicate monthly records by aggregating them by ['Loyalty#', 'Year', 'Month']. This summing approach was justified by our finding of perfect data continuity and preserved all valid activity. We also enforced the business rule *Points Accumulated = DistanceKM * 0.1* to correct inconsistent point values across this dataset.

4.2. Missing Values

After changing the data types, we did a further analysis on the missing values, after realizing there were 20 missing values on *Income* and *Customer Lifetime Value* and 14471 missing values on *Cancellation Date*.

On the missing values of the columns “*Income*” and “*Customer Lifetime Value*”, we observed that they all coincided on the same rows, which means that a row that contained a missing value on one of these columns, always contained a missing value on the other one. We also noticed that in these rows, the columns *Enrollment Date Opening* and *Cancellation Date* had the same date, which might explain why there were no values on *Income* and *Customer Lifetime Value*. In order to fill these missing values, we need to first check, after we merge the datasets, if these customers have flights or not, if not, we'll drop these rows, as they are not really customers of the company. If these customer have flights, we'll check the flights dates and correct the enrollment date. Only after this procedure, we can fill the missing values, which we be filled with a central tendency measure(e.g. median, mean) of the rows that are really similar to this ones, which means, with the mean or median of rows that have the same city, marital status, loyalty status, etc.

About the 14471 missing values of *Customer Lifetime Value*, these missing values will not be filled, as it would be filling it with false information, as a customer can join the program and not cancel it. This column will be transformed in the future, so that the missing values will correspond to 0 which means they will not be an issue.

For the flights dataset, we corrected 5,858 records where flight counts were missing (NumFlights = 0) despite recorded distance. This was fixed via a dual-strategy imputation: new customers received a value based on the global average distance per flight, while existing customers received one based on their own flight history.

4.3. Feature Engineering

In the customers dataset, the first feature we created was “*rejoined_program*”, that had a “1” in every row that where the cancelation date was before the enrollment data. Afterwards, we featured engineered “*Days_in_prog*” that counted the number of days since the customer joined the program. Then we proceeded to create “*Cancelled_program*”, where we captured the clients with a cancelation date superior to the enrollment date. Finally, we created “*Enrollment_year*” and “*Enrollment_month*”, in which we retained the year and the month, respectively, the customer enrolled in the program. With

these variables, we can keep important insights from the date variables and not lose information when we have to delete the date time variables, as the model cannot extract information from them.

To prepare the data for clustering, we aggregated 36 months of flight records for each customer into a single static profile (e.g., *"TotalFlights"*), which also resolved data anomalies from 2019. During analysis, we found perfect (1.0) correlations between *"DistanceKM"* and *"PointsAccumulated"*, and between *"PointsRedeemed"* and *"DollarCostPointsRedeemed"*. To prevent multicollinearity (duplicate weighting in the model), we decided to drop the *"PointsAccumulated"* and *"DollarCostPointsRedeemed"* variables.

As next steps, we plan to create a *"Flights_per_day"* variable to, along with CLV, calculate the *"Average_spent_per_flight"* for each customer.

4.4. Outliers

About the outliers in customer dataset, we verified that *"Longitude"* and *"Income"* contained no outliers, *"Latitude"* contained only one outlier, but an outlier that we plan to keep as it represents real demographic data about our customers, while *"Customer Lifetime Value"* revealed to be a really right-skewed variable, containing a lot of outliers, that need to be carefully analysed and treated as they give us a really important insight about customer relevance. We plan to apply a log transformation to it, so that we keep the same distribution, without having to change any values and reducing the impact of the skewed values.

Finally, our statistical review of the flights dataset's key behavioral features (e.g., TotalFlights, TotalDistance) revealed they were highly skewed. Clustering algorithms like K-Means are sensitive to such distributions and to features on different scales. Therefore, our final preprocessing step involves applying a log1p transformation to these skewed features to reduce the impact of outliers and normalize their distributions. This is followed by applying a StandardScaler to ensure all features contribute equally to the distance calculations in the clustering model.

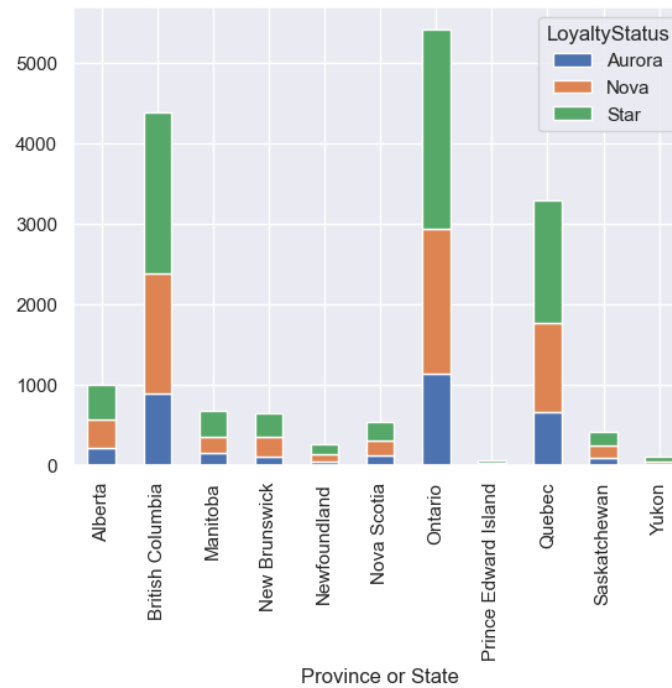
5. CONCLUSION

In short, our Exploratory Data Analysis (EDA) directly was a guide to further in-depth preprocessing steps. During this initial phase, we identified that by combining duplicates, filling in missing flight counts, removing redundant features and transforming skewed data, we could turn the raw, customer and flight records into a clean, static and normalized dataset. This careful preparation is key to ensuring our clustering model uses meaningful and well scaled features, which will ultimately lead to more accurate and interpretable customer segments for Amazing Airlines International marketing campaigns, that will be the focus of our future work.

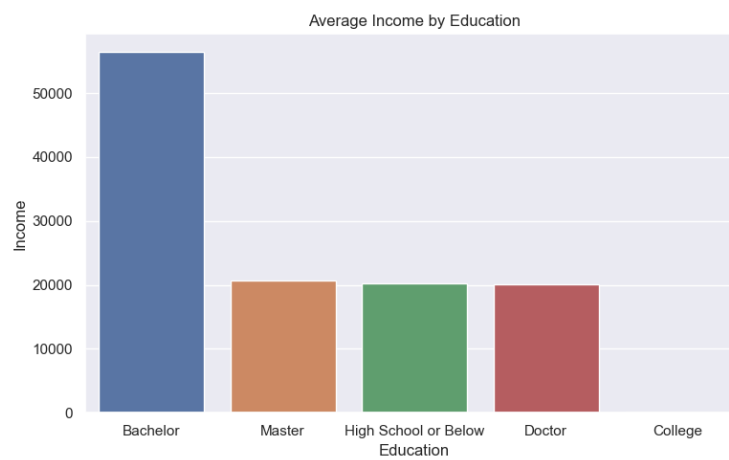
6. APPENDIX A

Appendix A contains relevant visualizations from AIAI's Customer Dataset.

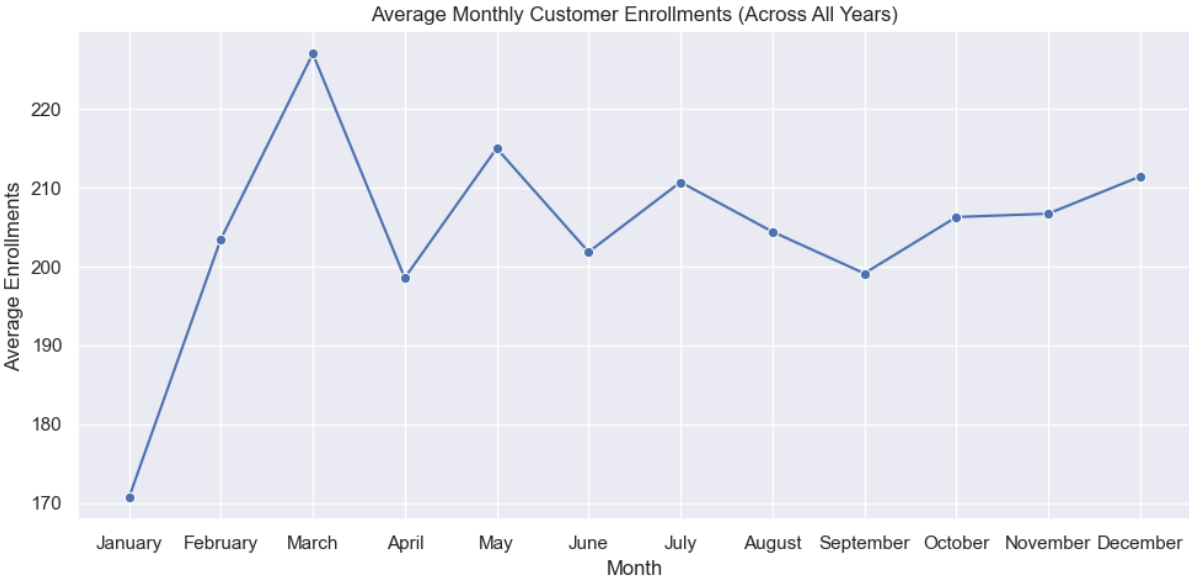
Graphic 1: Loyalty Status type per Province/State



Graphic 2: Average Income by Education level



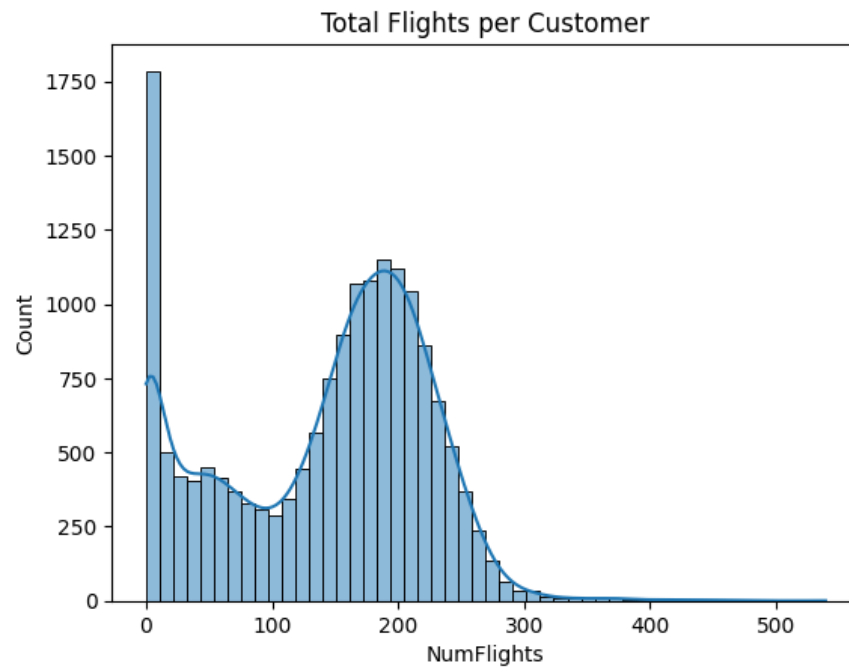
Graphic 3: Average Monthly Customer Enrollments (2019-2021)



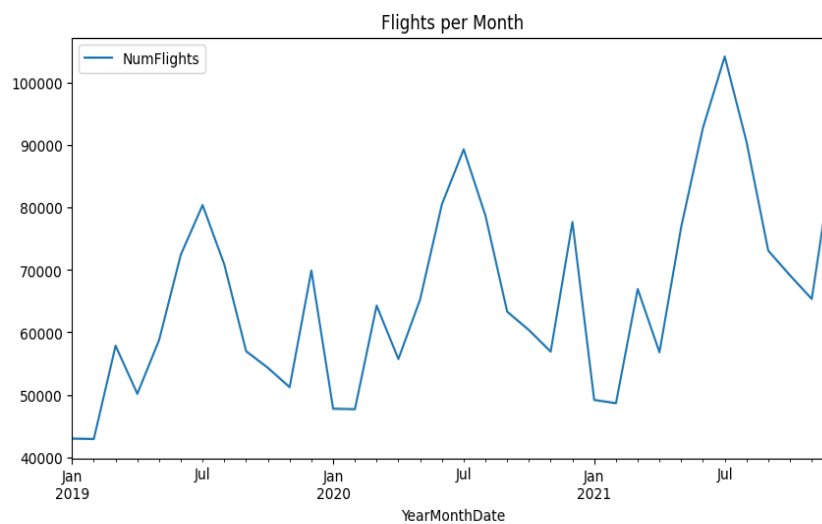
7. APPENDIX B

Appendix B contains relevant visualizations from AIAI's Flights Dataset.

Graphic 4: Total flights per Customer



Graphic 5: Trend of Flights per Month



8. ANNEX

8.1. Annex AI Usage Statement

AI tools were used for coding syntax assistance and writing correctness assessment review (Gemini) and report proofreading (Grammarly for grammar checks).

All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking (ideation, code assistance, refining, proofreading).

8.2. Annex Contribution Statement

Individual student contributions to deliverables.

André Ferreira (Student ID: 20250398)

- GitHub management
- Code: Flights Dataset
- Report: Business Understanding, Data Analysis of Flights Dataset, Conclusion

Fausto Gomes (Student ID: 20221915)

- Code: Flights Dataset
- Report: Results of Flights Dataset
- Poster: design and writing

Maria Francisca Gonçalves (Student ID: 20221942)

- Report: Abstract, CRISP-DM Methodology, Data Analysis of Customer Dataset
- Poster: design and writing
- Click-Up management (project management tool)

Miguel Matos (Student ID: 20221925)

- Code: Customer Dataset
- Report: Results of Customer Dataset

8.3. Annex Responsibility Statement

We, the group members listed above, confirm that the handout was developed equitably by all members, considering the code development, report and poster.

Moreover, we certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

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